

Coding & Solution

Date	15 March 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

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The **Credit Card Approval Prediction** project is developed using Machine Learning and Web technologies to predict whether a credit card application should be approved or rejected. The solution includes data preprocessing, model training, prediction, and a user-friendly web interface. The code follows modular programming practices, making it easy to maintain, test, and extend.

Solution Summary

Field	Details
Repository Link / URL	https://ai-ml-smartbridge-project-credit-card.onrender.com
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Bootstrap
Key Features Implemented	User login, Applicant data input form, Data preprocessing, Credit card approval prediction using Machine Learning, Prediction result display, Admin dashboard, Input validation
Pending / Incomplete Features	Email notification, Explainable AI (prediction reason), Advanced analytics dashboard, Cloud deployment
Setup / Run Instructions	1. Install Python 3.x. 2. Install required libraries using <code>pip install -r requirements.txt</code>. 3. Run the Flask application using <code>python app.py</code>. 4. Open the browser and visit <code>http://localhost:5000</code>.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The project follows a modular architecture with separate components for frontend, backend, machine learning model, and dataset.
- Input validation and exception handling improve system reliability.
- The machine learning model is trained using historical credit card application data to provide accurate approval predictions.
- The solution can be enhanced by integrating cloud deployment, real-time database support, and Explainable AI (XAI) features for better decision transparency.